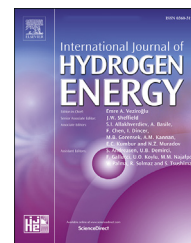


Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/hydro

How to choose endothermic process for thermochemical waste-heat recuperation?

Dmitry Pashchenko

Samara State Technical University, 244 Molodogvardeiskaya Str., Samara, 443100, Russia

ARTICLE INFO

Article history:

Received 5 February 2020

Accepted 30 April 2020

Available online 26 May 2020

Keywords:

Thermochemical recuperation

Hydrocarbons

Gibbs free energy minimization

Recuperation rate

Heat transformation coefficient

ABSTRACT

The thermochemical waste-heat recuperation (TCR) systems by steam reforming of various hydrocarbon fuels are considered. A method for determining the TCR systems efficiency is proposed. The methodology is based on the determination of the heat recuperation rate and heat transformation coefficient. TCR due to steam reforming of methanol, ethanol, glycerol, propane, and methane was analyzed. To obtain the initial data for energy analysis of TCR systems, the thermodynamic analysis was performed. With the help of Aspen HYSYS was determined the synthesis gas composition and reaction enthalpy for all investigated variants. The investigation was performed for a wide temperature range from 400 to 1200 K, for the steam-to-fuel ratio of 1, and pressures of 1 bar. It was established that TCR due to the steam reforming of ethanol, glycerol, and propane in the temperature range above 800 K, it is possible to achieve complete recuperation of the exhaust after the furnace. As a results of investigation, the practical recommendations were given for choosing endothermic reaction for the thermochemical waste-heat recuperation systems: in the temperature range up to 600 K for TCR it is necessary to choose a methanol steam reforming reaction; in the temperature range from 600 to 1000 K - the reaction of steam reforming of ethanol and glycerol; in the temperature range above 1000 K - the reaction of steam reforming propane and methane.

© 2020 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

Introduction

Industrialization is largely dependent on hydrocarbon fuel consumption. Today, transportation systems, heat and power generation, and industrial sectors are heavily dependent on hydrocarbon fuel and depletion of non-renewable resources as well as the increasing rate of greenhouse gas generation has become one the main concerns [1,2]. The limited hydrocarbon fuel resources call for increasing energy efficiency of fuel-consuming equipment. This is due to the fact that lower fuel consumption while maintaining fuel-consuming

equipment performance leads to a decrease in greenhouse gas emissions.

The low energy efficiency of various fuel-consuming plants is usually caused by a large loss of heat with flue gases, heat loss through the wall and other heat losses [3,4]. Traditional methods of increasing energy efficiency are based on the recovery of the exhaust heat. One of the most common methods of heat recovery is to heat the air for combustion by the exhaust heat [5,6]. In addition, thermochemical heat recuperation as a way for energy efficiency increasing has been widely discussed recently [7–9]. The essence of thermochemical heat recuperation (TCR) of exhaust flue gases is the use of exhaust heat to perform the endothermic chemical

E-mail address: pashchenkodmitry@mail.ru.

<https://doi.org/10.1016/j.ijhydene.2020.04.279>

0360-3199/© 2020 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights reserved.

reactions such as for example hydrocarbons reforming with steam or carbon dioxide.

For thermochemical heat recuperation, various endothermic reactions can be used. The TCR systems by hydrocarbons reforming can be divided into two blocks depending on the type of fuel used. The first is thermochemical heat recuperation due to liquid fuel reforming [10,11]; the second is thermochemical recuperation due to reforming of gaseous fuels [9,12]. As the oxidizing agent of fuels, steam is widely used [13]. However, there are suggestions for using combustion products as an oxidizing agent, a mixture of combustion products and steam, carbon dioxide [14], etc.

The results of using thermochemical heat recuperation show the big potential of such technology to increase the energy efficiency of fuel-consuming plants. Moreover, TCR can be considered as a promising way for increasing the environmental performance efficiency of fuel-consuming equipment [15]. For example, Popov et al. published the results of the investigation of various schemes of thermochemical waste-heat recuperation in the industrial furnaces [9]. The authors considered the TCR systems using the methane reforming: (1) with pure steam; (2) with the recirculated flue gases (the components of flue gases) are considered to be the oxidant for methane reforming: steam (H_2O) and carbon dioxide (CO_2); (3) with mixture of steam and flue gases. Analysis of research shown, that if recirculated flue gases are used as an oxidant for the methane reforming, then the dependence of the methane consumption and the costs of primary energy on the volume fraction of recirculated flue gases used for the reforming is non-monotonic and has a minimum. Also, the authors reported the thermochemical recuperation system can help to save up to 10–25% of energy without affecting other parameters of the process.

Verkhivker and Kravchenko presented the results of the thermodynamic calculation of a combined-cycle plant with thermochemical recuperation using steam methane reforming. The authors determined that the total energy efficiency of a combined cycle plant with thermochemical recuperation exhaust heat after a steam generator has about 80–90%. It is possible because heat recuperation in TCR is not limited by isentropic expansion (Carnot efficiency). Moreover, they showed that thermochemical recuperation has raised the amount of gas fuel used in energy and chemical industries and has decreased the amount of waste carbon dioxide ejected into the environment by nearly 20% [13].

Gaber and co-workers [7] presented the results of the experimental study of a thermochemical heat recuperation process for utilizing the waste heat of oxy-fuel furnaces, with three significant modifications similar to those used in Popov's article [9]. The main efforts of the authors were powered to a determination of the temperature range and gas hourly space velocity where thermochemical recuperation has maximum efficiency. The authors reported that the efficiency of thermochemical recuperation using depends on the methane conversion while the methane conversion depends on temperature, steam-to-methane ratio, space velocity.

In the above-mentioned studies, the methane reforming reaction was used as the endothermic reaction. The authors noted that the maximum efficiency of the TCR system due to

methane reforming is in the temperature range above 700–800°C.

An extensive discussion of the use of liquid hydrocarbons, in particular, ethanol and methanol, was found in the works of Tartakovsky and his coauthors. For example, the article “Fuel reforming in internal combustion” engines [8] shows a comprehensive study of fuel reforming using in the internal combustion engines. Tartakovsky and Sheintuch [8] presented a historical perspective of thermochemical recuperation using, a discussion on the considerations to be made before choosing a primary fuel for reforming purposes, and the main processes in fuel reforming. Moreover, Poran et al. presented the results of an experimental investigation of a direct injection internal combustion engine with high-pressure thermochemical recuperation [15]. The authors reported not only a significant improvement in the indicated efficiency of about 19–30% but also a reduction in NO_x, CO, HC, and CO₂ emissions by up to 97, 91, 96 and 15%, respectively. This paper and few others [12] present a high potential of the TCR concept not only from the waste heat recovery but also from the numerous advantages of hydrogen combustion.

Chakravarthy et al. [16] presented the thermodynamic analysis of the internal combustion engine with thermochemical heat recuperation due to the reforming of various liquid fuels. The authors demonstrated an increase in the energy efficiency of ICE with thermochemical recuperation with comparison ICE without thermochemical recuperation. Moreover, Gaber et al. reported the results of thermochemical analysis and experimental investigation of a recuperative waste heat recovery system for the tri-reforming of light oil [11]. The authors experimentally investigated the efficiency of thermochemical recuperation on temperature.

In these works on thermochemical recuperation due to liquid fuel reforming, the authors argue that TCR has high efficiency in the temperature range above 400–500°C. At the same time, TCR due to the methanol reforming is already effective at a temperature of 150–200°C, while for TCR due to the glycerol reforming, maximum efficiency occurs at temperatures above 800°C [10].

The papers shown above examine various endothermic reactions in a wide range of technological parameters. At the same time, it is rather difficult to compare the TCR systems due to various chemical reactions because there is no common methodology for the selection of initial fuels for thermochemical recuperation. Therefore, this work aims to develop a methodology for assessing the efficiency of using various chemical reactions for thermochemical heat recuperation, as well as a comparative analysis of the efficiency of using various fuels for TCR. In other words, this article answers the question: “How to choose an endothermic process for thermochemical waste-heat recuperation?” and shows the results of determining the effectiveness of the TCR systems due to various endothermic processes.

However, one of the main criteria for choosing the endothermic reaction for thermochemical recuperation is the availability of hydrocarbon fuel and the oxidizing agent. In other words, fuel and oxidizing agent should have low cost and high availability, as well as the thermochemical recuperation system, should have high efficiency. Therefore, for

further analysis, it is considered that all components for endothermic reactions have the same availability.

Methodology

TCR concept

The basic concept of thermochemical waste-heat recuperation is the preliminary endothermic transformation of the initial fuel by reforming reaction. For example, such a transformation is possible in the process of reforming hydrocarbons. To carry out such a process, heat supply is necessary because the reforming reaction is usually endothermic. The heat deficit for the reforming reaction in the TCR systems is covered by the use of secondary energy resources (waste heat), for example, the heat of the exhaust flue gases. As a result of such an endothermic transformation, synthetic gas is formed – hydrogen-rich gas fuel. This hydrogen-rich gas fuel has a total combustion heat greater than the combustion heat of the original fuel. A schematic diagram of an industrial furnace with thermochemical waste-heat recuperation is shown in Fig. 1.

In the traditional industrial gas furnaces, the chemical energy of the fuel is converted into heat in one stage by direct combustion. In the gas furnaces with thermochemical recuperation, the process of converting the chemical energy of fuel into heat is divided into two stages. The first stage is the endothermic process of the initial fuel reforming using the heat of the exhaust flue gases. The second stage is the combustion of reaction products, i.e. converted fuel (hydrogen-rich gas fuel). Such fuel has a higher combustion heat than it for the original hydrocarbon fuel. In addition, a comparative analysis of the combustion processes of primary and converted fuels shows that the environmental performances in the latter case are also better [15].

Analyzing the schematic view in Fig. 1, it can be concluded that for TCR it is possible to use various fuels and various oxidizing agents, in the interaction of which an endothermic reaction takes place. However, when TCR is used on an industrial scale, the cost of the initial fuel and oxidizing agent

should be minimal. For this reason, in industry for TCR, mainly steam is used as an oxidizing agent.

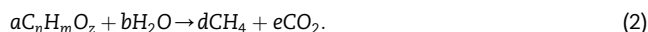
For comparative analysis in this article, the following hydrocarbon fuels were selected as the initial fuel: methanol, ethanol, glycerol, methane, propane, isooctane. Steam is selected as an oxidizing agent. In each case, the amount of steam added to reformer is equal to the amount required to completely steam reform the fuel to carbon monoxide and hydrogen. It should be mentioned that the steam-to-fuel ratio is important for the reforming reaction. However, an increase in the ratio above the optimum may have a negative energy effect [17].

It should be noted that the catalytic steam reforming of hydrocarbons in thermal effect and the amount of hydrogen produced is several times greater than non-catalytic endothermic processes such as pyrolysis, cracking and depolymerization of hydrocarbons. Therefore, endothermic hydrocarbon decomposition reactions, as a rule, are not considered for thermochemical recuperation.

The specific type of steam reforming reaction of hydrocarbons depends on the process conditions (temperature, pressure, steam-to-fuel ratio, etc.). For example, at high temperatures ($T > 1000^\circ\text{C}$), reactions practically proceed to form only hydrogen and carbon monoxide (high-temperature reforming):



At low temperatures ($T > 500\text{K}$), the yield of products is strongly biased towards the formation of methane and carbon dioxide, and the resulting gasification reaction can be represented as (low-temperature reforming):



In the general case, reactions (1) and (2) are accompanied by one more independent reversible reaction – water-gas shift reaction:



Unfortunately, the hydrocarbon reforming processes is complicated by undesirable reactions of the coke formation, which leads to carbonization and poisoning of the catalysts.

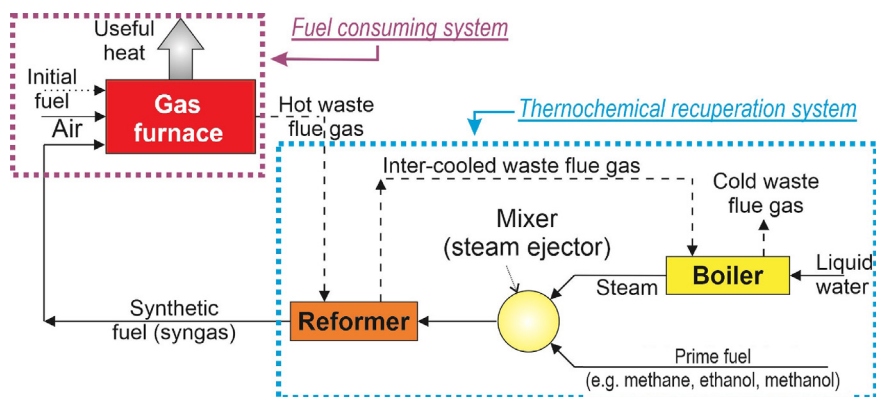


Fig. 1 – Schematic view of an industrial gas furnace with thermochemical recuperation of the exhaust gas heat by steam reforming of hydrocarbons.

Coke, deposited on the surface of the catalyst, deactivates it, blocking active centers; clogging the pores leads to the destruction of the catalyst due to differences in the expansion coefficients of materials. Coke formation in the thermochemical recuperation process can be controlled by following actions [18]:

- an appropriate choice of source fuel;
- catalyst promotion;
- variation of steam-to-fuel ratio;
- measures for the rational organization of the fuel reforming process (for example, optimal catalyst shape, elimination of local temperature inhomogeneities, etc.);
- the use of special fuel additives;
- catalyst regeneration.

Methodology

To evaluate the effectiveness of using various chemical reactions for thermochemical recuperation, the heat recovery rate (HRR) can be used as an efficiency criterion. The heat recovery rate indicates the ratio of the recovered heat to the heat of the exhaust flue gases. In other words, HRR indicates how much heat of the exhaust gas was recovered to the heat cycle of the industrial furnace. Analyzing the schematic diagram in Fig. 1, it can be seen that the heat recuperation of the exhaust flue gases is carried out in two elements – a thermochemical reformer and a steam generator (boiler). Therefore, according to the definition, HRR can be expressed as follows:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{H_{rec}^{ref} + H_{rec}^{st.gen}}{H_{exh}}, \quad (4)$$

where H_{rec} – a recovered heat in the TCR system, kJ; H_{rec}^{ref} , $H_{rec}^{st.gen}$ – a recovered heat in a reformer and a steam generator, respectively; H_{exh} – an exhaust heat, kJ.

It can be seen from expression (4) that HRR is determined not only by the recovered heat but also by the heat of the exhaust flue gases. The efficiency of the application of the TCR systems by steam reforming of hydrocarbons is depending on the heat transferred in the reformer for the reforming reaction and synthesis gas heating. If methane reforming doesn't occur in the reformer, then the efficiency of using the TCR system equals zero or even has a negative effect, due to the fact that exhaust heat is used for the generation of steam.

The enthalpy of the exhaust gases depends on the steam-to-fuel ratio in the reformer. The enthalpy of the exhaust gases for 1 mol of inlet fuel (at excess air ratio equal to unity) can be determined as follows:

$$H_{exh} = [n_{CO_2} \cdot c_{\mu, CO_2} + n_{N_2} \cdot c_{\mu, N_2} + (n_{H_2O} + \beta) \cdot c_{\mu, H_2O}] \cdot T_{exh}, \quad (5)$$

where n_i is the number of moles of i -th element of the combustion products, mol; $c_{\mu, i}$ is the molar heat capacity, J/(mol°C); $\beta = H_2O^{in}/Fuel^{in}$ is steam-to-fuel ratio at the reformer inlet.

The recovered heat in the reformer depends on the heat transferred in the reformer for the reforming reaction and

synthesis gas heating. Therefore, the recovered heat in the reformer can be calculated as follows:

$$H_{rec}^{ref} = H_{sfr} + \Delta H_{sg}, \quad (6)$$

where H_{sfr} – enthalpy of steam fuel reforming process, kJ; ΔH_{sg} – increasing enthalpy of syngas, kJ.

Here in eq. (6), the enthalpy of the steam fuel reforming process can be calculated depends on the process variables such as temperature, pressure, steam-to-fuel ratios, type of catalyst, residence time, etc. This study uses the equilibrium enthalpy of the steam fuel reforming process, which is determined using thermodynamic analysis.

In the expression for the heat recovery rate (4), there is the term of recovered heat in the steam generator. $H_{rec}^{st.gen}$ can be calculated as follows:

$$H_{rec}^{st.gen} = \Delta Q_{st.gen} = H_{exh}' - H_{exh}'' = (h_{steam} - c_p \cdot t_w^{in}) G_{steam}^{in}, \quad (7)$$

where H_{exh}' , H_{exh}'' is enthalpy of exhaust gas before and after a steam generator, kJ; h_{steam} is the enthalpy of steam after a steam generator, kJ; G_{steam}^{in} is the mass flow rate of water before a steam generator, kg/s; t_w^{in} is temperature of feed water, °C; c_p is the heat capacity of feed water, kJ/(kg°C).

To determine the initial data for expressions (4)–(7), it is necessary to know the composition of the synthesis gas after reformer as well as the enthalpy of the reforming process. In this study, thermodynamic analysis is used for these purposes.

An algorithm of a heat recovery rate calculation for the TCR systems can be presented via a control flow chart presented in Fig. 2.

Thermodynamic analysis

The equilibrium synthesis gas composition, fuel conversion, and enthalpy of the fuel reforming process was calculated using thermodynamic analysis. The methodology of a thermodynamic analysis by Gibbs free energy minimization (GFEM) method was discussed in the author's previous publications [19,20] and it was also introduced in the details by other authors [21–23]. Thermodynamic analysis of hydrocarbon reforming with steam using GFEM technique was performed to calculate the equilibrium composition of the syngas. GFEM method is based on the concept that the analyzed thermochemical system is thermodynamically favourable if total Gibbs free energy has the lowest value and differential of Gibbs free energy is zero for each processing conditions [24]:

$$dG^t = 0 \quad (8)$$

where G^t is total Gibbs free energy.

At moderate low pressure (before 5–10 bar) and high temperature, the system can be considered as ideal [25]. Also, the synthesis gas after the fuel reformer is calculated, based on the following admissions:

- steady flow process with the negligible change of kinetic and potential energies;

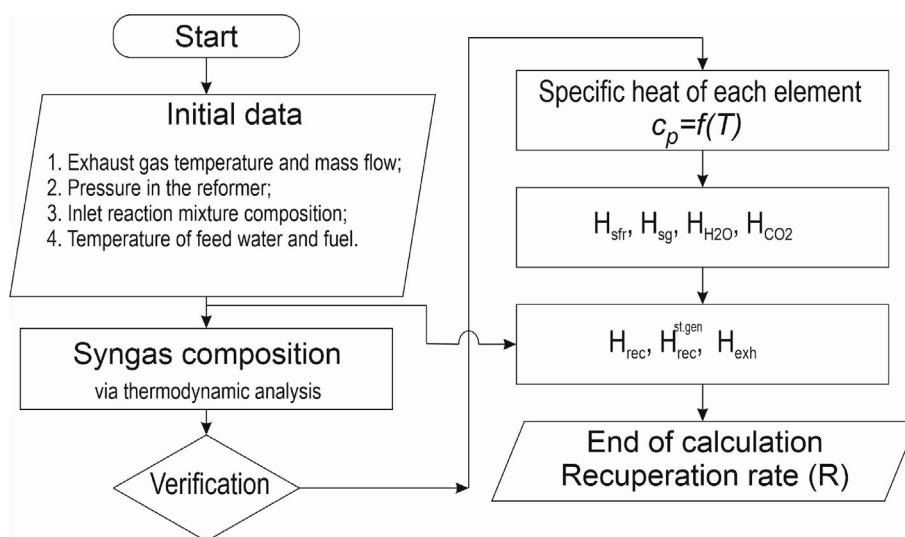


Fig. 2 – An algorithm of a heat recovery rate calculation for the TCR systems.

- perfect insulation of reformer with negligible heat loss;
- chemical reactions with known reactants and product compositions are equilibrium.

Total Gibbs free energy of a hydrocarbon reforming system is determined by the sum of the chemical potential of all chemical components:

$$G^t = \sum n_i \mu_i \quad (9)$$

where n_i is number of moles of species i ; μ_i is chemical potential of species i .

The chemical potential of each element is determined as follow:

$$\mu_i = \Delta G_{fi}^0 + RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (10)$$

where G_{fi}^0 is standard Gibbs free energy of formation of species i ; R is molar gas constant; T is temperature of the system; f_i is fugacity of pure species i ; f_i^0 is fugacity of pure species i at the standard state.

If combine the expressions above, the equation for total Gibbs free energy of the chemical system can be written as:

$$G^t = \sum n_i \Delta G_{fi}^0 = \sum n_i RT \ln \left(\frac{f_i}{f_i^0} \right) \quad (11)$$

The thermodynamic analysis based on the Gibbs free energy minimization method (GFEM) was done via special software Aspen-HYSYS [26–29]. This program is a powerful process simulator for modeling steady-state process systems where chemical reactions take place. Aspen-HYSYS is capable to simulate a single-phase or multiphase system at equilibrium composition. A Gibbs reactor in this software is used to determine the composition of synthesis gas at the reformer outlet in reaction system under the conditions of Gibbs free energy minimization. The Gibbs reactor allows calculating the composition of synthesis gas by Gibbs free energy minimization of each component in the reaction system according to

the reaction network. Fig. 3 shows the computational scheme of the thermochemical recuperation process for Aspen HYSYS.

Carbon formation and deposition are other factors that have to be discussed especially at low steam-to-fuel ratio and moderately high temperatures. However, there is sufficient information in the literature that carbon formation can be controlled by improving the coking resistance of the catalysts as well as by regulating the operating conditions such as steam-to-carbon and fuel-to-oxidant ratio. In this study, the oxygen-to-carbon ratio in all different feed gas compositions is either equal to or greater than unity, which indicates that there is enough oxygen to oxidize and prevent the formation of solid carbon. Hence, carbon formation was omitted from equilibrium expressions.

Results and discussion

Verification

The efficiency of the TCR systems is determined by technological parameters, such as temperature, pressure and steam-to-fuel ratio because such parameters determine the

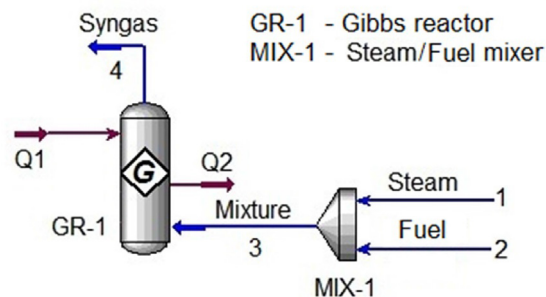


Fig. 3 – The computational scheme of a thermochemical recuperation process for Aspen HYSYS.

composition of the synthesis gas and the process enthalpy. The energy efficiency of the TCR systems is determined based on the results of thermodynamic analysis. To verify the results of the thermodynamic analysis, the results obtained by the author using Aspen HYSYS were compared with the results of other authors. As mentioned above, the main task of thermodynamic analysis is to determine the enthalpy of the fuel reforming process, as well as determining the equilibrium composition of the synthesis gas at the reformer outlet. The reaction enthalpy and synthesis gas composition is the basis for the calculation of heat recuperation rate. In this study, the author does not set the main goal to show the results of the thermodynamic analysis of the hydrocarbon reforming process. Such results are widely represented in the scientific literature, for example, [30]. The main goal of the thermodynamic analysis presented by the author is to obtain generalized results for the calculation of the energy efficiency of TCR systems. To verify the results of the thermodynamic analysis, the obtained data were compared with the results of thermodynamic analysis and experimental investigation of other authors.

Fig. 4a shows a comparison of the results of an experimental study of ethanol steam reforming over a Ni-based catalyst [31] with results of the thermodynamic analysis. Biswas and Kunzru showed that the composition of the catalyst has a significant effect on the synthesis gas composition. The experiments were performed at high residence time (contact time or space velocity) about $4.5 \text{ g}_{\text{cat}}/\text{h}/\text{mol}$. Such a high time allows us to expect a close to the equilibrium composition of reformat products. Moreover, Fig. 4b presents the comparison of the results of thermodynamic analysis of the steam methanol reforming process performed by the author via Aspen HYSYS with the results of the thermodynamic analysis performed by Yu et al. via HSC Chemistry 5.11 [32]. An obvious correlation of the results is observed for both steam reforming processes of ethanol and methanol.

To verify the results of the thermodynamic analysis of the steam methane reforming process, the syngas compositions were compared with another thermodynamic study of

Verkhivker and Kravchenko [13]. Thermodynamic calculations were performed for different temperatures, pressures, and inlet reaction mixture compositions. A comparison of the results of the synfuel composition is shown in Table 1. The upper results are the mole fraction of each component that was calculated with the help of IVTANTHERMO, the middle results are the mole fraction that was obtained with Aspen-HYSYS, the lower results are the mole fraction that was taken from Verkhivker's work [13], in which an analytical calculation of the steam methane reforming was done. For all the results obtained, good convergence is observed.

The comparison of the thermodynamic analysis results with the results of other authors [33,34] showed a good correlation for the wide range of the operational conditions.

Heat recuperation rate

As shown above, the heat recuperation of the exhaust heat in the TCR system is achieved in two elements – a steam generator (boiler) and a reformer (according to Fig. 1). In the steam generator, heat recuperation occurs due to steam heating, which, after reaction with methane, turns into synthesis gas. Thus, a traditional heat recuperation by heating the combustion components with exhaust heat takes place in the steam generator. In the reformer, heat recuperation occurs according to the thermochemical principle – heating the reaction mixture and transformation of the heat of the reaction mixture into chemical energy (due to the endothermic reforming reaction). Therefore, in order to evaluate the efficiency of using TCR systems, various options for heat recuperation were considered.

Fig. 5 shows the dependence of the heat recuperation rate due only to the chemical transformation of exhaust heat in the reformer on temperature of exhaust gases at a pressure of 1 bar and the steam-to-fuel ratio of 1. In this case, the HRR is defined as the ratio of the process enthalpy and the exhaust gas enthalpy (according to eq. (4)). When determining HRR in the reformer, it is assumed that the temperature of the steam-fuel mixture at the reformer inlet is equal to the temperature

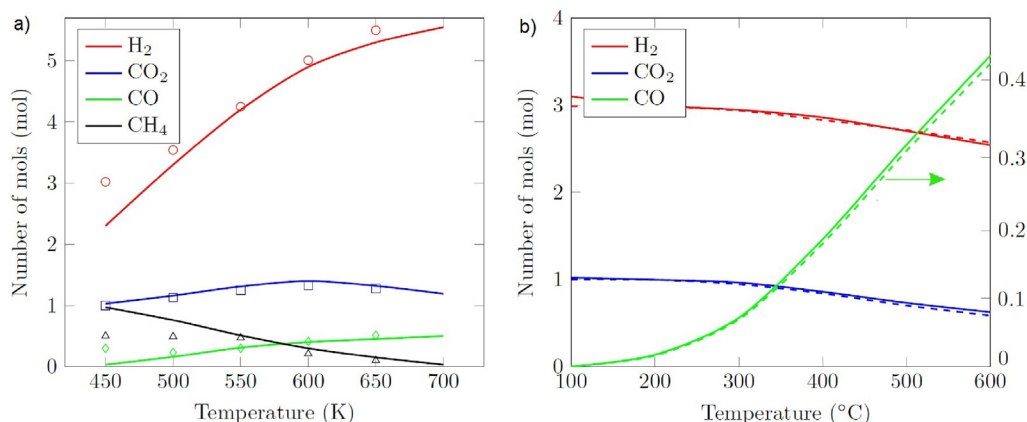
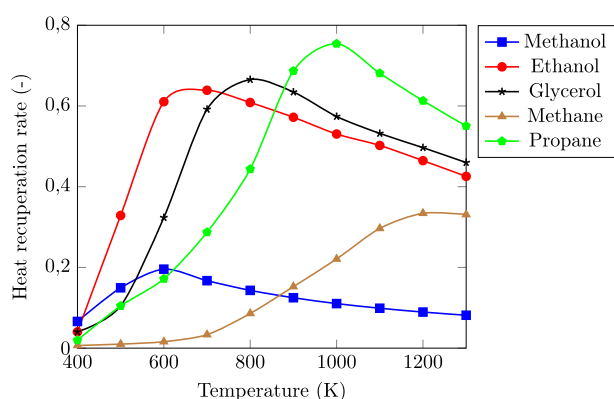


Fig. 4 – Comparison the results of thermodynamic analysis with: a) experimental data [31] on ethanol steam reforming over a Ni-based catalyst (hollow points are experimental results, solid lines – results of author's thermodynamic analysis); b) thermodynamic study of methanol steam reforming [32] (dashed line – the results from Refs. [32]; solid lines – results of author's thermodynamic analysis).

Table 1 – The comparison of the synfuel gas composition, % (mol/mol).

Temperature	Pressure	Source	Gas composition, %				
			H ₂	CO	H ₂ O	CO ₂	CH ₄
1000 K	1 bar	IVTAN	63.01	14.32	16.32	5.23	1.12
		ASPEN	63.40	14.38	16.12	5.08	1.02
		Verkhivker [13]	63.42	14.43	16.11	5.12	0.92
1000 K	5 bar	IVTAN	60.66	14.82	17.32	5.55	1.65
		ASPEN	61.60	14.98	17.92	4.08	1.42
		Verkhivker [13]	61.44	14.93	17.97	4.18	1.48
1100 K	1 bar	IVTAN	63.88	16.08	16.02	3.93	0.09
		ASPEN	63.65	16.14	16.15	3.98	0.08
		Verkhivker [13]	63.76	16.11	16.11	3.94	0.08
1100 K	5 bar	IVTAN	61.28	15.02	17.87	4.24	1.59
		ASPEN	61.57	14.99	17.88	4.11	1.45
		Verkhivker [13]	61.45	14.93	17.96	4.18	1.48

**Fig. 5 – Heat recuperation rate achieved in the reformer only by chemical recuperation.**

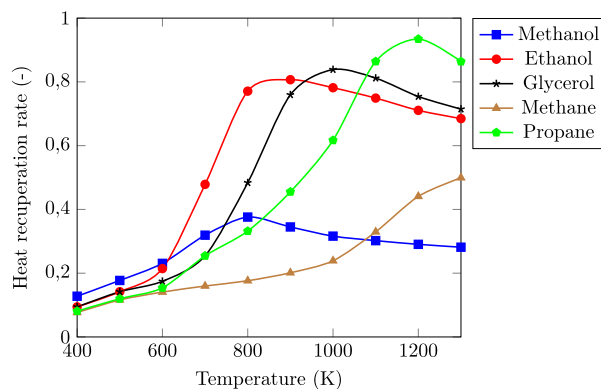
of the synthesis gas at the outlet and the temperature of the process is equal to the temperature of the exhaust gases. In other words, Fig. 5 shows the dependence of HRR, which is achieved through chemical recuperation – the transformation of the exhaust gases enthalpy into the chemical energy of a new synthetic fuel.

Fig. 5 shows that for all the studied endothermic reactions, HRR in the reformer first increases, then reaches its maximum value, and then decreases. An increase in HRR is due to an increase in the enthalpy of the reforming process. With increasing process temperature, the enthalpy of the process increases. However, the process enthalpy reaches its maximum value and increases slightly above a certain temperature. Therefore, after reaching the maximum value with increasing temperature, HRR decreases, because denominator – the heat of the exhaust gases increases with a practically constant enthalpy of the process. HRR in the reformer has the maximum value for the steam reforming reaction of propane. This is due to the fact that the process enthalpy is about 500 kJ/mol at a temperature of about 1000 K. At the same time, the HRR in the reformer is minimal for TCR system due to steam reforming of methanol. This is due to the fact that the maximum enthalpy of this process is about 50 kJ/mol. For steam methane reforming, the HRR in the reformer increases monotonously and reaches a maximum at a temperature of about 1200 K.

In the real reformers, the temperature of the synthesis gas at the outlet of the reformer (for example, steam reforming) is 150–200 °C lower than the temperature of the heating gases. Therefore, in Fig. 6 shows the temperature dependence of HRR, provided that the temperature of the synthesis gas is 200 °C lower than the temperature of the exhaust gas. Moreover, to unify the initial conditions, it is assumed that the temperature of the mixture of steam and fuel before the reformer is 200 °C. In this case, the heat recuperation in the reformer is determined as follows:

$$H_{rec}^{ref} = H_{sfr} + \Delta H_{sg}. \quad (12)$$

Fig. 6 shows the change in heat recuperation rate due to the chemical transformation of exhaust heat and heating of the synthesis gas as a function of temperature. Fig. 6 makes it clear that in the temperature range up to 600 K, the maximum HRR occurs in TCR systems due to the steam reforming of methanol. However, with increasing temperature, HRR in TCR due to steam reforming of methanol increases weaker than HRR of the TCR system due to steam reforming of ethanol, glycerol, and propane. At the same time, the temperature range up to 1100 K HRR has a minimum value for the TCR system due to steam reforming of methane. Fig. 6 shows that HRR in the reformer can reach 0.8–0.95 for TCR due to steam reforming of ethanol, glycerol, and propane. In other words, it

**Fig. 6 – Heat recuperation rate achieved in the reformer by thermochemical recuperation – chemical transformation of exhaust heat and synthesis gas heating.**

can be argued that in the reformer due to the above-mentioned reactions, it is possible to recuperate almost all the heat of the exhaust flue gases.

However, according to expression (4), heat recuperation in the TCR systems occurs in two elements – a reformer and a steam generator. According to the scheme in Fig. 1, steam generation for thermochemical recuperation occurs through the use of the heat of the exhaust flue gases. However, for TCR systems in which there is HRR of about 1, there may be a deficit in the heat of the exhaust flue gases. In other words, the heat of the exhaust gases after the reformer will not remain enough heat for steam generation. In this case, a shortage (deficit) of heat occurs. Fig. 7 shows HRR in the entire TCR system as a function of temperature, as well as the dependence of the heat deficit in the TCR system on temperature. For the dependence in Fig. 7 the following initial data were used: pressure 1 bar, steam-to-fuel ratio 1, water and fuel temperature at system inlet is 300 K, synthesis gas temperature after the reformer is 200 K lower than the temperature of the exhaust gases.

It can be seen from Fig. 7 that for TCR due to the steam reforming of ethanol, glycerol and propane at temperatures above 750 K, 850 K, 1030 K, respectively, a heat deficit occurs. This means that the heat of the exhaust gas is not enough to complete the thermochemical recuperation process for given conditions. Therefore, to cover the heat deficit, it is necessary to use an external heat source, for example, to take heat from the furnace. However, on the other hand, Fig. 7 shows the possibility of achieving complete recuperation of the heat of the exhaust flue gases after the furnace. Thus, it becomes possible useful to use all the heat of the exhaust flue gas.

The most balanced TCR systems are due to steam reforming of methanol at temperatures below 1000 K and steam reforming of methane at temperatures above 1000 K. For such thermochemical recuperation systems, the heat of the exhaust gas is enough to carry out the process in the given parameters.

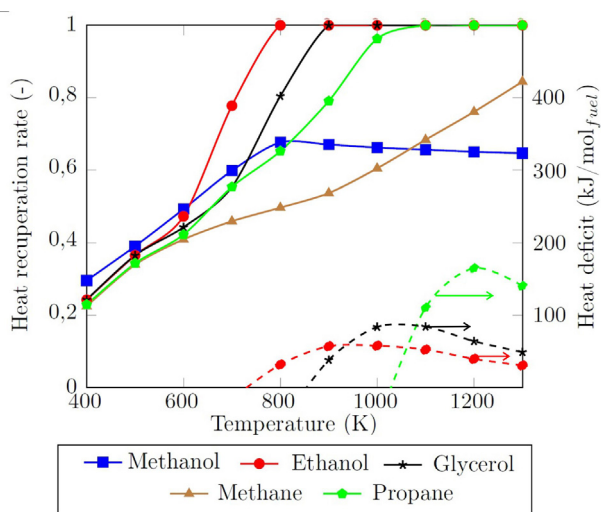


Fig. 7 – Heat recuperation rate achieved in the reformer and steam generator and heat deficit in the TCR systems.

Despite the obvious advantages of using HRR as an indicator of the effectiveness of TCR systems, this indicator does not evaluate the quality of the synthesis gas (hydrogen-rich fuel) after the reformer. To assess the efficiency of the transformation of the exhaust heat into the chemical energy of new synthetic fuel, the heat transformation coefficient is used.

Heat transformation coefficient

Initial fuels before reforming have different the combustion heat (low heat value – LHV). Therefore, to assess the fuel efficiency in a thermochemical recuperation system, it is necessary to know how many times the LHV of new synthetic fuel is greater than the LHV of the initial fuel. The ratio of LHV of new synthetic fuel to LHV of the original fuel is called the heat transformation coefficient. The heat transformation coefficient (HTC) can be expressed as follows:

$$\eta_q = \text{LHV}_{\text{synfuel}} / \text{LHV}_{\text{fuel}}. \quad (13)$$

If the reforming reaction is endothermic, then the heat transformation coefficient is greater than unity. Fig. 8 shows a comparison of the heat transformation coefficients for various fuels at different temperatures.

Fig. 8 shows the increase in temperature of exhaust gases increases the low heat value of the synthesis gas in comparison to initial fuels. This is because with an increase in the temperature of the process, the reaction enthalpy increases. Fig. 8 depicts TCR due to steam methane reforming has a maximum heat transformation coefficient for the entire investigated temperature range from 400 to 1200 K. The maximum heat transformation coefficient is 1.243 for steam methane reforming. For the temperature range below 800 K the maximum heat transformation coefficient is for TCR due to steam ethanol reforming.

The HTC for TCR due to steam propane reforming has high values only in the temperature range above 1000 K. This is due to the fact that propane has a high low heat value and, in

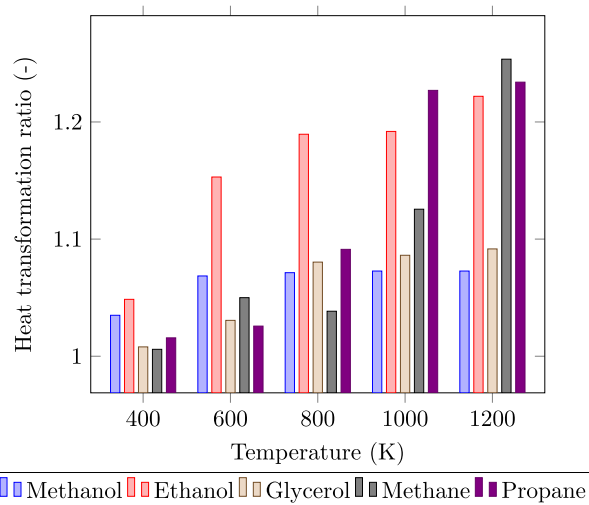


Fig. 8 – The transformation coefficients for various fuels vs. temperature.

expression (13), the denominator does not allow reaching a high η_q value.

Conclusion

Two criteria for the effectiveness of the thermochemical recuperation (TCR) systems have been proposed – the heat recuperation rate (HRR) and the heat transformation coefficient (HTC). The methods for calculating HRR and HTC is proposed. A comparative analysis of the efficiency of using various fuels in the thermochemical recuperation systems due to steam reforming of methanol, ethanol, glycerol, propane and methane has been performed. It has been established that in the temperature range up to 600 K, the maximum HRR has TCR due to steam reforming of methanol. For TCR due to the steam reforming of ethanol, glycerol and propane in the temperature range above 800 K, it is possible to achieve complete recuperation of the exhaust after the furnace. At the same time, the aforementioned TCR systems may cause a deficit in the exhaust heat. The most balanced TCR in the temperature range up to 800 K is TCR due to steam reforming of methanol, in the temperature range above 800 K is TCR due to steam reforming of methane.

The use of TCR leads to a change in the fuel composition used to produce thermal energy in a furnace. The heat transfer coefficient (HTC) shows how many times the low heat value of a new synthetic fuel is greater than the low heat value of the initial fuel. HTC has maximum value for TCR due to steam reforming of methane.

Thus, the question “How to choose endothermic process for thermochemical waste-heat recuperation?” has the following answers:

- initial fuel for TCR should have greater availability and low cost;
- heat recuperation rate should tend to unity;
- heat transformation coefficient should have a maximum value.

As practical recommendations, the following suggestions can be distinguished: in the temperature range up to 600 K for TCR it is necessary to choose a methanol steam reforming reaction; in the temperature range from 600 to 1000 K - the reaction of steam reforming of ethanol and glycerol; in the temperature range above 1000 K - the reaction of steam reforming propane and methane.

Acknowledgment

This work is supported by the Russian Science Foundation under grant 19-19-00327.

Nomenclature

R	heat recuperation rate
H_i	enthalpy of i-th flow
n_i	number of moles of i-th element

β	steam-to-fuel ratio
μ_i	chemical potential of element i
G^t	value of total Gibbs free energy
G_{fi}^0	standard Gibbs free energy of formation of element i
R	molar gas constant
T	temperature
f_i	fugacity of pure elements i
f_i^0	fugacity of pure elements i at the standard state
C_{p_i}	specific heat of i component
LHV	low heating value
HRR	heat recuperation rate
HTC	heat transfer coefficient
TCR	thermochemical recuperation

REFERENCES

- [1] Balitskiy S, Bilan Y, Strielkowski W, Štreimikienė D. Energy efficiency and natural gas consumption in the context of economic development in the European Union. *Renew Sustain Energy Rev* 2016;55:156–68.
- [2] Clark DE, Dybiz C, Hanson A, Nourzad F. No shut-off policies and natural gas consumption. *Resour Energy Econ* 2017;48:19–29.
- [3] Zhang Q, Zhao X, Lu H, Ni T, Li Y. Waste energy recovery and energy efficiency improvement in China's iron and steel industry. *Appl Energy* 2017;191:502–20.
- [4] Miró L, Gasia J, Cabeza LF. Thermal energy storage (tes) for industrial waste heat (iw) recovery: a review. *Appl Energy* 2016;179:284–301.
- [5] Mokhtar M, Burns S, Ross D, Hunt I. Exploring multi-objective trade-offs in the design space of a waste heat recovery system. *Appl Energy* 2017;195:114–24.
- [6] Yin Q, Du W-J, Ji X-L, Cheng L. Optimization design and economic analyses of heat recovery exchangers on rotary kilns. *Appl Energy* 2016;180:743–56.
- [7] Gaber C, Demuth M, Prieler R, Schluckner C, Hochenauer C. An experimental study of a thermochemical regeneration waste heat recovery process using a reformer unit. *Energy* 2018;155:381–91.
- [8] Tartakovsky L, Sheintuch M. Fuel reforming in internal combustion engines. *Prog Energy Combust Sci* 2018;67:88–114.
- [9] Popov S, Svistunov I, Garyaev A, Serikov E, Temyrkanova E. The use of thermochemical recuperation in an industrial plant. *Energy* 2017;127:44–51.
- [10] Pashchenko D, Gnutikova M, Karpilov I. Comparison study of thermochemical waste-heat recuperation by steam reforming of liquid biofuels. *Int J Hydrogen Energy* 2019;45:4174–81.
- [11] Gaber C, Demuth M, Schluckner C, Hochenauer C. Thermochemical analysis and experimental investigation of a recuperative waste heat recovery system for the tri-reforming of light oil. *Energy Convers Manag* 2019;195:302–12.
- [12] Pan F, Cheng X, Wu X, Wang X, Luo P. Transient performances of the gas turbine recuperating waste heat through hydrogen rich fuels. *Int J Hydrogen Energy* 2019;44(56):29743–51.
- [13] Verkhivker G, Kravchenko V. The use of chemical recuperation of heat in a power plant. *Energy* 2004;29(3):379–88.
- [14] Pashchenko D. Thermochemical waste-heat recuperation by steam methane reforming with flue gas addition. *Int J Energy Res* 2019;43(6):2216–26. <https://doi.org/10.1002/er.4436>.

- [15] Poran A, Thawko A, Eyal A, Tartakovsky L. Direct injection internal combustion engine with high-pressure thermochemical recuperation—experimental study of the first prototype. *Int J Hydrogen Energy* 2018;43(27):11969–80.
- [16] Chakravarthy VK, Daw CS, Pihl JA, Conklin JC. Study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines. *Energy Fuel* 2010;24(3):1529–37.
- [17] Pashchenko D. First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming. *Energy* 2018;143:478–87.
- [18] Korabelnikov A, Kuranov A. Usage of catalytic processes in advanced power plants of hydrocarbon-fueled hypersonic vehicles. In: 15th AIAA international space planes and hypersonic systems and technologies conference; 2008. p. 2571.
- [19] Pashchenko D. Thermochemical recovery of heat contained in flue gases by means of bioethanol conversion. *Therm Eng* 2013;60(6):438–43.
- [20] Pashchenko D. Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation. *Int J Hydrogen Energy* 2017;42(22):14926–35.
- [21] Zeng G, Tian Y, Li Y. Thermodynamic analysis of hydrogen production for fuel cell via oxidative steam reforming of propane. *Int J Hydrogen Energy* 2010;35(13):6726–37.
- [22] Adhikari S, Fernando S, Gwaltney SR, To SF, Bricka RM, Steele PH, Haryanto A. A thermodynamic analysis of hydrogen production by steam reforming of glycerol. *Int J Hydrogen Energy* 2007;32(14):2875–80.
- [23] Nahar G, Madhani S. Thermodynamics of hydrogen production by the steam reforming of butanol: analysis of inorganic gases and light hydrocarbons. *Int J Hydrogen Energy* 2010;35(1):98–109.
- [24] Aktaş S, Karakaya M, Avcı AK. Thermodynamic analysis of steam assisted conversions of bio-oil components to synthesis gas. *Int J Hydrogen Energy* 2009;34(4):1752–9.
- [25] Chan S, Wang H. Thermodynamic analysis of natural-gas fuel processing for fuel cell applications. *Int J Hydrogen Energy* 2000;25(5):441–9.
- [26] Øi LE. Aspen hysys simulation of co₂ removal by amine absorption from a gas based power plant. In: The 48th scandinavian conference on simulation and modeling (SIMS 2007). Linköping University Electronic Press; 2007. p. 73–81. Göteborg (Säro), no. 027.
- [27] Zahid U, Al Rowaili FN, Ayodeji MK, Ahmed U. Simulation and parametric analysis of co₂ capture from natural gas using diglycolamine. *Int J Greenhouse Gas Control* 2017;57:42–51.
- [28] Feyzi V, Beheshti M, Kharaji AG. Exergy analysis: a co₂ removal plant using a-mdea as the solvent. *Energy* 2017;118:77–84.
- [29] Pham TN, Long NVD, Lee S, Lee M. Enhancement of single mixed refrigerant natural gas liquefaction process through process knowledge inspired optimization and modification. *Appl Therm Eng* 2017;110:1230–9.
- [30] Sharma YC, Kumar A, Prasad R, Upadhyay SN. Ethanol steam reforming for hydrogen production: latest and effective catalyst modification strategies to minimize carbonaceous deactivation. *Renew Sustain Energy Rev* 2017;74:89–103.
- [31] Biswas P, Kunzru D. Steam reforming of ethanol for production of hydrogen over ni/ceo₂–zro₂ catalyst: effect of support and metal loading. *Int J Hydrogen Energy* 2007;32(8):969–80.
- [32] Yu KMK, Tong W, West A, Cheung K, Li T, Smith G, Guo Y, Tsang SCE. Non-syngas direct steam reforming of methanol to hydrogen and carbon dioxide at low temperature. *Nat Commun* 2012;3(1):1–7.
- [33] Tabrizi FF, Mousavi SAHS, Atashi H. Thermodynamic analysis of steam reforming of methane with statistical approaches. *Energy Convers Manag* 2015;103:1065–77.
- [34] Lutz AE, Bradshaw RW, Keller JO, Witmer DE. Thermodynamic analysis of hydrogen production by steam reforming. *Int J Hydrogen Energy* 2003;28(2):159–67.